

// Day 14 of 30

VISUAL EXPLAINER

What Happens When You Call an API

→ Swipe through all 6 slides



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Step 1: DNS Resolution

Browser finds the server

You type: `api.example.com/users/1`

Browser asks DNS:

'Where is `api.example.com`?'

DNS responds:

'`203.0.113.42`'

Now browser knows the IP address.

💡 DNS = phonebook of the internet



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Step 2: TCP + TLS Handshake

Establishing a secure connection

TCP 3-way handshake:

Client → SYN → Server

Server → SYN-ACK → Client

Client → ACK → Server

TLS (HTTPS):

→ Exchange certificates

→ Agree on encryption

→ Secure channel ready

💡 This takes ~50-100ms. Happens once per connection.



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Step 3: HTTP Request Sent

Your request hits the wire

```
GET /users/1 HTTP/1.1
Host: api.example.com
Authorization: Bearer eyJhbG...
Content-Type: application/json
Accept: application/json
```

💡 Headers carry auth, format, and metadata



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Step 4: Load Balancer → Server

Traffic routing

Request hits Load Balancer

- Checks which server is healthy
- Routes to Server 2 (least busy)

Server 2 receives request

- Nginx/Gateway passes to Spring Boot
- DispatcherServlet takes over

💡 Load balancer = traffic cop for your servers



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Step 5: Spring Boot Processing

Controller → Service → Repository

DispatcherServlet

- Filter Chain (auth, CORS)
- @GetMapping("/users/{id}")
- UserController.getUser(1)
- UserService.findById(1)
- UserRepository.findById(1)
- Hibernate → SQL → MySQL
- Result → DTO → JSON

💡 This is where YOUR code lives



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Step 6: Response Back

Server → Client in milliseconds

```
HTTP/1.1 200 OK  
Content-Type: application/json
```

```
{  
  "id": 1,  
  "name": "Shubham",  
  "role": "Backend Engineer"  
}
```

```
Total time: ~120ms  
DNS: 20ms, TCP: 30ms  
Server: 50ms, Transfer: 20ms
```

💡 Understanding this = understanding backend engineering



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done = true

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